

Explainable Hybrid Learning for Cardiovascular Risk Prediction with mRMR–PCA Feature Optimization

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Abstract—Cardiovascular disease (CVD) remains one of the leading causes of death worldwide, thereby creating a need for risk predictions which are accurate and interpretable. Herein, this work proposes a robust hybrid machine learning framework to predict cardiovascular risk. The framework integrates minimum Redundancy Maximum Relevance (mRMR) feature selection, Principal Component Analysis (PCA), and an explainable multi-model learning strategy. We evaluate the architecture on the Framingham Heart Study dataset comprising 4,240 patient records with mixed clinical and demographic information. A total of seven supervised learning classifiers—Extra Trees, Random Forest, Gradient Boosting, CatBoost, XGBoost, Decision Tree, and Multilayer Perceptron—have been evaluated for performance using stratified 10-fold cross-validation and test validation. To address feature redundancy and class imbalance, mRMR, PCA, and SMOTE were employed. The best-performing model, which combines PCA with Extra Trees, achieves 98.56% accuracy, 97.81% precision, 99.35% recall, 98.57% F1-score, and a ROC–AUC score of 0.9988. The use of explainable AI tools has also been on the rise as a means of improving transparency. Both global and local explanations, which employ SHAP and LIME explanations, identify important risk factors for clinical conditions such as age, blood pressure, and blood glucose. Overall, the model has a good balance of predictive performance, robustness, and interpretability, indicating that the model is appropriate for application in clinical practice.

Index Terms—Cardiovascular Risk Prediction, Explainable AI, Feature Selection, Dimensionality Reduction, PCA, mRMR, SHAP, LIME, Clinical Decision Support

I. INTRODUCTION

Cardiovascular disease (CVD) is the leading cause of death worldwide, imposing a significant strain on global health-care systems. The World Health Organization reports that cardiovascular problems cause millions of deaths each year, emphasizing the need for precise risk prediction tools for early intervention and preventative care [10]. Effective cardiovascular risk assessment is critical for improving patient outcomes and facilitating informed treatment decisions.

The availability of large-scale clinical and epidemiological datasets has greatly aided data-driven cardiovascular risk analysis. Benchmark datasets, like the Framingham Heart Study and Cleveland Heart Disease dataset, give extensive demographic, lifestyle, and clinical features, allowing for effective application of machine learning (ML) approaches [8], [9]. ML techniques are better adapted than standard statistical models

to capture the complex and non-linear interactions found in cardiovascular data.

However, ML-based models frequently suffer from high-dimensional feature spaces and duplication among clinical variables, which can impair predictive performance and interpretation. To solve these challenges, feature selection and dimensionality reduction techniques such as minimal Redundancy Maximum Relevance (mRMR) and Principal Component Analysis (PCA) are increasingly important [2], [6]. Furthermore, the increasing complexity of predictive models has highlighted the need for Explainable Artificial Intelligence (XAI) techniques, such as SHAP and LIME, to ensure transparency and clinical trust [5], [7].

Figure 1 shows the overall workflow of the proposed robust hybrid framework for cardiovascular risk prediction.

Motivated by the constraints of existing methodologies, this paper provides a strong hybrid framework that combines mRMR-based feature selection with PCA-driven dimensionality reduction, analyzes numerous machine learning classifiers, and adds explainable AI techniques. The suggested methodology seeks to strike a balance between predictive accuracy, computational efficiency, and interpretability, resulting in reliable and clinically relevant cardiovascular risk prediction.

II. LITERATURE REVIEW

Cardiovascular disease (CVD) remains the leading cause of mortality worldwide, emphasizing the need for accurate and early risk prediction systems [10]. Widely used clinical datasets, including the Framingham Heart Study, enable data-driven modeling of cardiovascular risk factors [8].

Conventional statistical approaches such as Logistic Regression offer interpretability but often fail to capture complex nonlinear relationships in clinical data. Consequently, machine learning methods have gained significant attention, with studies on the Framingham dataset reporting strong performance from models such as Random Forest and XGBoost [1], [2].

Nevertheless, cardiovascular datasets frequently suffer from high dimensionality and redundant clinical attributes. Feature optimization techniques, including mRMR and PCA, have demonstrated improvements in model efficiency and stability [3], [6]. Although deep learning architectures like CNN–BiLSTM provide promising results, their computational

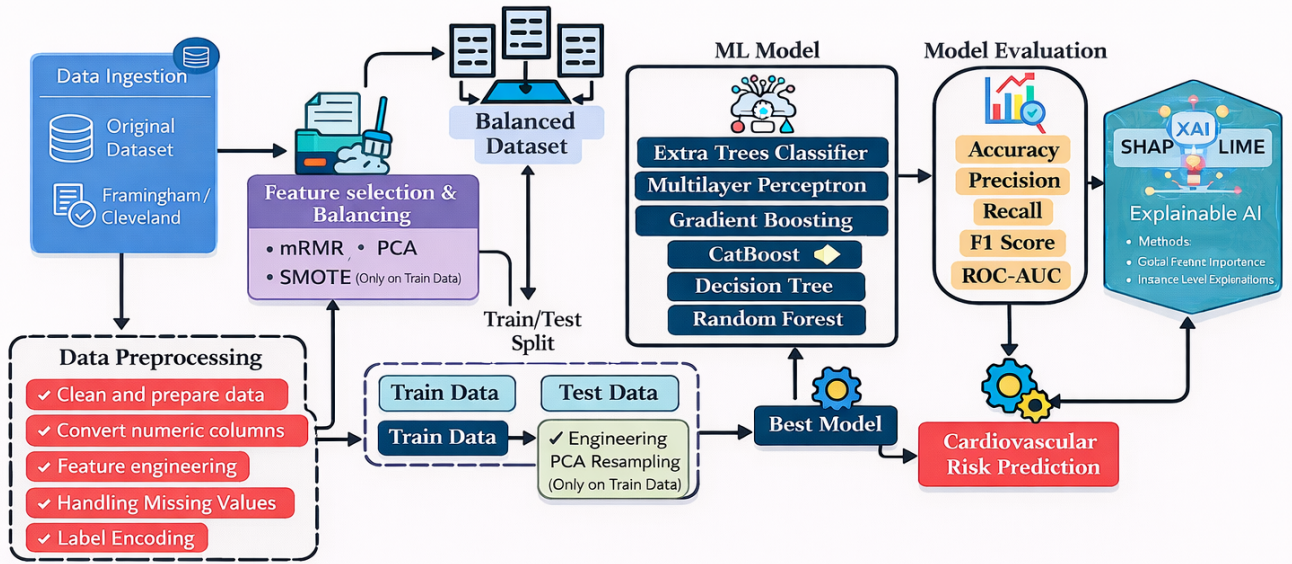


Fig. 1: End-to-end overview of the proposed robust hybrid methodology for cardiovascular risk prediction.

complexity and limited interpretability restrict clinical applicability [4].

To enhance transparency, Explainable Artificial Intelligence (XAI) methods such as SHAP and LIME are increasingly incorporated, consistently identifying clinically meaningful predictors like age, blood pressure, and glucose-related variables [5], [7].

Despite these advancements, many studies rely on isolated optimization techniques or post-hoc explanations. This highlights the need for integrated frameworks combining effective feature selection, diverse model learning, and inherent interpretability. Additionally, benchmark datasets such as the Cleveland Heart Disease dataset remain important for comparative evaluation across clinical cohorts [9].

III. METHODOLOGY

This section outlines the method used in the study. It covers the dataset description, data preparation and splitting, feature selection, class balance, hyperparameter optimization, model construction, and performance evaluation.

A. Dataset Description

The proposed framework is evaluated using the Framingham Heart Study dataset, containing demographic, lifestyle, and clinical data from 4,240 patients with 15 predictive features and a binary cardiovascular outcome. The dataset is widely used and clinically validated for cardiovascular risk modeling, supporting reliable comparative evaluation. For methodological consistency, findings are also discussed alongside studies using the Cleveland Heart Disease dataset [1], [5], [8], [9]. Its longitudinal structure and well-defined clinical attributes further enable reproducible supervised learning and benchmarking.

B. Data Preprocessing

Data preparation is the foundation of robust machine learning, and it ensures the data is maintained in a clean and usable form. In the case of missing values in numerical data, we use the median to fill the gap. In the case of categorical variables, we use the most frequent category to fill the gap. All the categories will then be converted to numbers, and the numeric variables will undergo z-score normalization.

To increase the predictive capabilities, we calculate derived variables that are more meaningful from a clinical perspective. For improving the speed and efficiency, we use feature scaling and reduce the dimensions.

Evaluation of the model employs stratified 10-fold cross-validation to maintain the same distribution of classes as the original dataset, with an equal proportion of both positive and negative instances used for training and evaluation sets.

1) *Class Imbalance Analysis*: The original Framingham dataset shows a significant class imbalance between low-risk (Class 0) and high-risk (Class 1) cardiovascular illnesses. Figures 2, Fig. 3, and Fig. 4 illustrate that the majority class outnumbers the minority class, which can bias learning algorithms towards dominant patterns and diminish sensitivity for high-risk predictions.

This imbalance is especially important in clinical risk assessment, where accurately identifying high-risk patients is key.

2) *Balancing Techniques Using SMOTE*: To address the issue of imbalanced classes, SMOTE is employed to address class imbalance by generating synthetic minority class samples through interpolation rather than duplication. The SMOTE Technique uses interpolation to create new minority class samples. This will increase the minority class samples rather than simply replicating them.

TABLE I: 10-Fold Stratified Average Cross-Validation Results (Mean $\pm$ SD)

Model	Feature Selection	No. of Features	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Extra Trees	Original	18	0.9905 $\pm$ 0.0050	0.9868 $\pm$ 0.0079	0.9944 $\pm$ 0.0051	0.9906 $\pm$ 0.0050	0.9975 $\pm$ 0.0025
	mRMR	10	0.9848 $\pm$ 0.0035	0.9724 $\pm$ 0.0067	0.9981 $\pm$ 0.0018	0.9851 $\pm$ 0.0035	0.9995 $\pm$ 0.0006
	PCA	12	0.9953 $\pm$ 0.0023	0.9942 $\pm$ 0.0041	0.9964 $\pm$ 0.0031	0.9953 $\pm$ 0.0023	0.9990 $\pm$ 0.0008
Random Forest	Original	18	0.9773 $\pm$ 0.0055	0.9611 $\pm$ 0.0086	0.9950 $\pm$ 0.0051	0.9777 $\pm$ 0.0053	0.9974 $\pm$ 0.0028
	mRMR	10	0.9675 $\pm$ 0.0037	0.9410 $\pm$ 0.0073	0.9975 $\pm$ 0.0040	0.9684 $\pm$ 0.0035	0.9990 $\pm$ 0.0009
	PCA	12	0.9783 $\pm$ 0.0060	0.9617 $\pm$ 0.0103	0.9964 $\pm$ 0.0031	0.9787 $\pm$ 0.0058	0.9990 $\pm$ 0.0009
Gradient Boosting	Original	18	0.7328 $\pm$ 0.0163	0.7238 $\pm$ 0.0140	0.7528 $\pm$ 0.0293	0.7378 $\pm$ 0.0186	0.8055 $\pm$ 0.0221
	mRMR	10	0.7141 $\pm$ 0.0141	0.7178 $\pm$ 0.0122	0.7055 $\pm$ 0.0229	0.7115 $\pm$ 0.0163	0.7910 $\pm$ 0.0159
	PCA	12	0.7514 $\pm$ 0.0153	0.7377 $\pm$ 0.0173	0.7812 $\pm$ 0.0320	0.7584 $\pm$ 0.0167	0.8302 $\pm$ 0.0167
CatBoost	Original	18	0.8882 $\pm$ 0.0095	0.8565 $\pm$ 0.0120	0.9330 $\pm$ 0.0135	0.8930 $\pm$ 0.0091	0.9563 $\pm$ 0.0087
	mRMR	10	0.8437 $\pm$ 0.0137	0.8138 $\pm$ 0.0182	0.8921 $\pm$ 0.0140	0.8510 $\pm$ 0.0124	0.9207 $\pm$ 0.0102
	PCA	12	0.9185 $\pm$ 0.0103	0.8824 $\pm$ 0.0173	0.9664 $\pm$ 0.0112	0.9223 $\pm$ 0.0093	0.9753 $\pm$ 0.0054
XGBoost	Original	18	0.9353 $\pm$ 0.0098	0.8972 $\pm$ 0.0137	0.9836 $\pm$ 0.0082	0.9384 $\pm$ 0.0090	0.9841 $\pm$ 0.0066
	mRMR	10	0.9074 $\pm$ 0.0099	0.8630 $\pm$ 0.0143	0.9689 $\pm$ 0.0036	0.9128 $\pm$ 0.0088	0.9641 $\pm$ 0.0082
	PCA	12	0.9534 $\pm$ 0.0120	0.9197 $\pm$ 0.0204	0.9942 $\pm$ 0.0046	0.9554 $\pm$ 0.0109	0.9922 $\pm$ 0.0041
Decision Tree	Original	18	0.9223 $\pm$ 0.0056	0.8691 $\pm$ 0.0094	0.9944 $\pm$ 0.0048	0.9275 $\pm$ 0.0048	0.9223 $\pm$ 0.0056
	mRMR	10	0.9224 $\pm$ 0.0135	0.8673 $\pm$ 0.0212	0.9983 $\pm$ 0.0022	0.9280 $\pm$ 0.0117	0.9224 $\pm$ 0.0135
	PCA	12	0.9230 $\pm$ 0.0093	0.8692 $\pm$ 0.0139	0.9961 $\pm$ 0.0028	0.9283 $\pm$ 0.0081	0.9230 $\pm$ 0.0093
MLP	Original	18	0.5840 $\pm$ 0.0662	0.5765 $\pm$ 0.2374	0.6640 $\pm$ 0.3601	0.5428 $\pm$ 0.2607	0.7121 $\pm$ 0.0276
	mRMR	10	0.5477 $\pm$ 0.0430	0.5217 $\pm$ 0.2847	0.4492 $\pm$ 0.3896	0.3980 $\pm$ 0.2566	0.6496 $\pm$ 0.0750
	PCA	12	0.8244 $\pm$ 0.0193	0.8061 $\pm$ 0.0172	0.8546 $\pm$ 0.0327	0.8293 $\pm$ 0.0204	0.8891 $\pm$ 0.0151

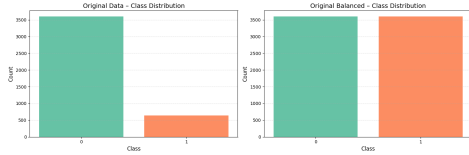


Fig. 2: Balanced class distribution after applying SMOTE on the original feature space.

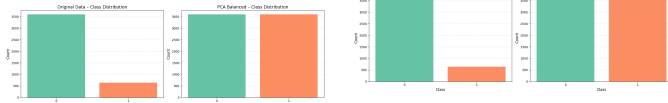


Fig. 3: Class distribution after applying SMOTE in the PCA-transformed feature space.

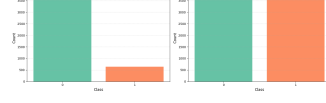


Fig. 4: Balanced class distribution after applying SMOTE on the mRMR-selected feature space.

To investigate how balancing at various stages of the pre-processing process affects performance, the SMOTE method is applied to three separate feature spaces: the original feature space, the space after PCA transformation, and the space formed by mRMR feature selection.

3) *SMOTE on Original Feature Space*: SMOTE is first applied to the original standardized feature set to establish a baseline balanced dataset, as illustrated in Fig. 2.

4) *SMOTE after PCA Transformation*: Next, SMOTE is applied in the PCA-transformed feature space, where dimensionality reduction captures dominant variance structures while suppressing noise, as shown in Fig. 3.

5) *SMOTE after mRMR Feature Selection*: Finally, SMOTE is performed after mRMR feature selection to balance the dataset using only the most relevant and non-redundant fea-

tures, as illustrated in Fig. 4.

This combined strategy enhances class balance while preserving feature relevance, contributing to improved model stability and generalization.

C. Feature Correlation Analysis

Correlation analysis is used to investigate inter-feature correlations and detect multicollinearity amongst clinical variables. Fig. 5 shows substantial correlations among blood pressure-related qualities, whereas interaction-based features also demonstrate redundancy with their constituent variables.

These findings support the use of feature selection and dimensionality reduction. Thus, mRMR is used to choose meaningful and non-redundant features, followed by PCA to project correlated variables onto an orthogonal feature space.

D. Feature Selection

A hybrid feature selection method is used to eliminate feature redundancy while also improving model performance. Minimum Redundancy The Maximum Relevance (mRMR) algorithm is first used to find features that are highly relevant to the target variable while minimizing inter-feature redundancy. Principal Component Analysis (PCA) is then used to turn the selected features into orthogonal components. A total of 12 principal components are retained, accounting for the majority of the variance in the data.

E. Data Balancing

The Framingham dataset's class imbalance is addressed with SMOTE to increase minority-class representation and prediction performance.

F. Hyperparameter Tuning

Hyperparameters are optimized using randomized search and stratified cross-validation, with optimal configurations chosen based on ROC–AUC performance (Table II).

TABLE II: Best Hyperparameters Identified for Evaluated Models

Model	Optimized Hyperparameters
Extra Trees	$n_estimators = 300, max_depth = 20$
Random Forest	$n_estimators = 200, max_features = \sqrt{}$
Gradient Boosting	$n_estimators = 150, learning_rate = 0.1, max_depth = 3$
CatBoost	$depth = 8, iterations = 500$
XGBoost	$learning_rate = 0.05, max_depth = 6$
Decision Tree	$max_depth = 10, min_samples_split = 5$
MLP	$hidden_layers = (100, 50), alpha = 0.001$

G. Machine Learning Models

Seven supervised machine learning classifiers are tested, including Extra Trees, Random Forest, Decision Tree, Gradient Boosting, XGBoost, CatBoost, and Multilayer Perceptron. These models cover a wide range of ensemble, tree, and neural network learning paradigms.

H. Performance Evaluation

Model performance is evaluated using accuracy, precision, recall, F1-score, and ROC–AUC. To assess robustness and stability, results are reported as mean and standard deviation across stratified cross-validation folds.

I. Overall Workflow

The proposed workflow comprises data preprocessing, correlation analysis, mRMR-based feature selection, PCA-driven dimensionality reduction, SMOTE-based balancing, model training with hyperparameter optimization, and explainability analysis. To avoid data leakage and overfitting, SMOTE, mRMR, and PCA are applied exclusively within the training folds of stratified cross-validation. Final performance is assessed using an independent hold-out test set to ensure unbiased generalization.

IV. RESULTS AND DISCUSSION

Tree-based models demonstrate strong performance compared to individual classifiers. As shown in Fig. 6, the Extra Trees classifier with PCA-based features yields fewer misclassifications, indicating improved classification stability.

Fig. 8 and Fig. 9 present the ROC and precision–recall comparisons, respectively. The PCA-transformed feature set achieves the highest ROC–AUC and maintains superior precision at higher recall levels, which is essential for reliable identification of high-risk cardiovascular cases.

Overall, the PCA-based Extra Trees model attains the best accuracy and ROC–AUC, demonstrating strong generalization and stable performance across stratified cross-validation folds.

Fig. 7 shows the learning curves of the Extra Trees classifier under original, mRMR-selected, and PCA-transformed feature

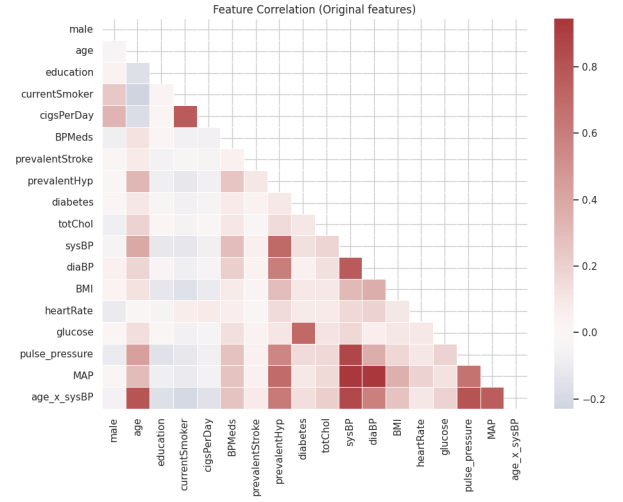


Fig. 5: Correlation heatmap of original clinical features illustrating inter-feature dependencies and multicollinearity.

TABLE III: Comparison with Similar Cardiovascular Risk Prediction Studies

Study	Dataset	Accuracy (%)	Methodological Highlights
Proposed Framework	Framingham Heart Study	98.56	Extra Trees classifier with PCA and mRMR feature selection, SMOTE-based class balancing, and SHAP–LIME explainability
Vasavi <i>et al.</i> [1]	Framingham Heart Study	94.2	Random Forest model with feature selection and explicit class imbalance handling
Ahmed <i>et al.</i> [2]	Framingham Heart Study	95.6	PCA-transformed features combined with XGBoost classifier and feature scaling
Roopini <i>et al.</i> [3]	Framingham Heart Study	85.11	Rough Set-based feature selection strategy with noise reduction techniques
Yuda <i>et al.</i> [5]	Framingham Heart Study	84.86	Logistic regression model using clinically derived attribute analysis
Yuda <i>et al.</i> [7]	Framingham Heart Study	95.9	XGBoost-based prediction model with SHAP-driven feature importance interpretation

sets. Validation accuracy increases steadily with training size across all configurations, indicating effective learning and improved generalization.

The PCA-based configuration exhibits the most stable convergence with a minimal training–validation gap, indicating reduced variance and improved robustness. In contrast, the original and mRMR feature sets show larger gaps at smaller sample sizes. Overall, PCA-driven dimensionality reduction enhances convergence and generalization of the Extra Trees classifier.

Fig. 10 presents the calibration curve of the Extra Trees classifier trained on PCA-transformed features. The dashed diagonal line indicates a perfectly calibrated model, while the solid curve represents the predicted probability distribution of

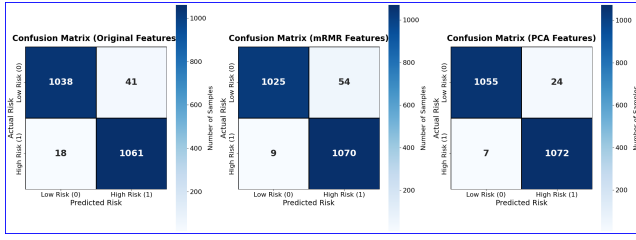


Fig. 6: Confusion matrix comparison of the Extra Trees classifier using original features, mRMR-selected features, and PCA-transformed features.

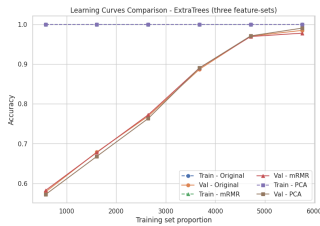


Fig. 7: Learning curve comparison of the Extra Trees classifier using original, mRMR-selected, and PCA-transformed feature sets.

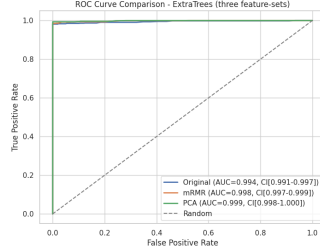


Fig. 8: ROC curve comparison of the Extra Trees classifier using original, mRMR-selected, and PCA-transformed feature sets.

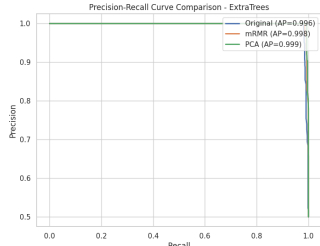


Fig. 9: Precision-Recall curve comparison of the Extra Trees classifier using original, mRMR-selected, and PCA-transformed feature sets.

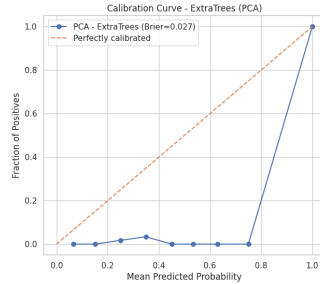


Fig. 10: Calibration curve of the Extra Trees classifier using PCA-transformed features. The dashed line represents perfect calibration.

the proposed model.

The calibration curve closely follows the ideal line at higher probability ranges, indicating dependable risk estimation for high-risk patients. Minor deviations at lower probabilities reflect slight underconfidence, commonly observed in highly discriminative models. The low Brier score further confirms strong probability calibration, supporting the framework's reliability for clinical decision support.

V. EXPLAINABLE ARTIFICIAL INTELLIGENCE FRAMEWORK

Explainability is incorporated using SHAP for global interpretation and LIME for local explanation. SHAP summary plots highlight the relative contribution of the 12 PCA components to cardiovascular risk prediction.

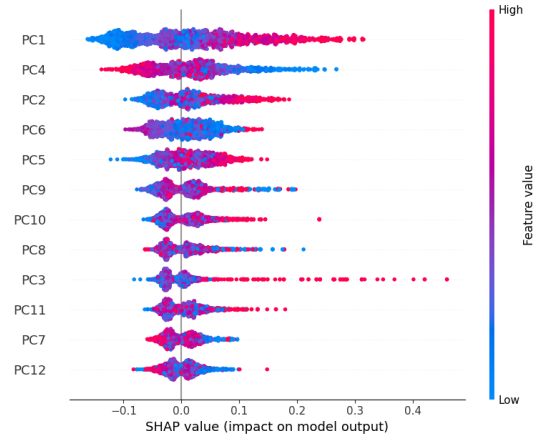


Fig. 11: SHAP summary plot of global importance for the 12 PCA components in the Extra Trees classifier.

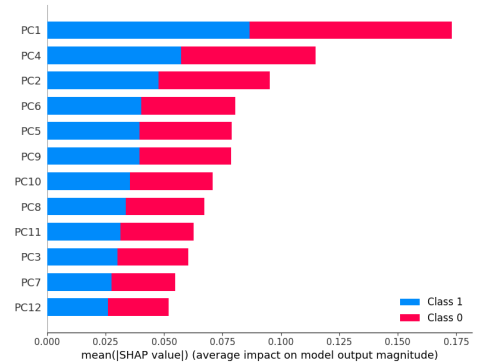


Fig. 12: Mean absolute SHAP-based global importance of PCA components for the Extra Trees classifier.

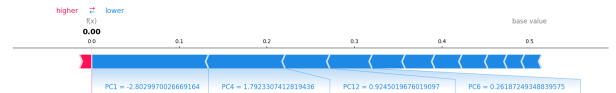


Fig. 13: Local SHAP explanation illustrating how PCA components increase or decrease cardiovascular risk for an individual prediction.

Fig. 12 illustrates the global importance of PCA components based on mean absolute SHAP values. PC1 and PC4 emerge as the most influential components, indicating their dominant contribution to cardiovascular risk prediction. The balanced contribution across both classes suggests stable decision behavior of the Extra Trees model after dimensionality reduction.

A. LIME-based Local Explanation of Model Prediction

Local Interpretable Model-agnostic Explanations (LIME) are employed to enhance model transparency by providing interpretable, instance-level explanations of individual predictions.

Figure 14 presents a local explanation for an individual test instance. Features such as absence of prevalent stroke, no blood pressure medication (BPMeds), and absence of diabetes

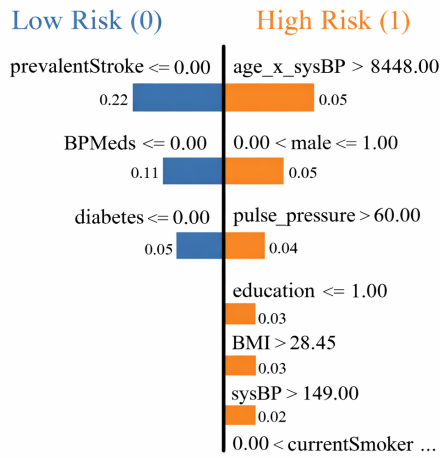


Fig. 14: LIME-based local explanation of an individual prediction, where blue bars indicate contributions to the Low-Risk class (0) and orange bars indicate contributions to the High-Risk class (1). Bar length reflects feature importance.

contribute toward the Low-Risk (class 0) prediction, while high systolic blood pressure, elevated BMI, pulse pressure, education level, and smoking status drive the prediction toward the High-Risk (class 1) class.

These results indicate that the model relies on clinically meaningful risk factors, and the use of LIME enhances interpretability by clearly explaining feature-level contributions to individual predictions.

VI. FUTURE WORK

While the recommended framework offers excellent predictive performance and interpretability, there are numerous opportunities for further enhancement. Future research will focus on improving the framework's robustness and fairness by testing it on varied ethnic and geographical populations.

Furthermore, merging longitudinal and real-time data from wearable sensors with IoT-based health monitoring systems could provide dynamic and continuous cardiovascular risk assessment. Further research could look into multimodal data fusion by including electronic health records, medical imaging, and physiological signals such as ECG to capture more subtle illness patterns.

Finally, converting the framework into a user-friendly clinical decision support system and conducting prospective real-world validation studies will be crucial in assessing its practical utility and impact on patient outcomes.

VII. CONCLUSION

This paper proposes a reliable and explainable hybrid machine learning framework for cardiovascular risk prediction. By combining minimum Redundancy Maximum Relevance (mRMR) and Principal Component Analysis (PCA) with multiple supervised classifiers, the framework effectively reduces feature redundancy while improving predictive performance. Evaluation on the Framingham Heart Study dataset using

stratified 10-fold cross-validation demonstrates that the PCA-based Extra Trees classifier outperformed other models in terms of accuracy and discrimination.

Beyond prediction performance, the methodology prioritizes interpretability by employing Explainable Artificial Intelligence techniques. Global and local explanations based on SHAP and LIME disclose clinically essential risk factors such as age, blood pressure, and glucose-related characteristics, increasing model transparency and clinician confidence.

Overall, the proposed approach achieves a balanced trade-off between accuracy, robustness, and interpretability, making it suitable for real-world cardiovascular risk assessment. Future work will focus on validation across diverse populations, leakage-free preprocessing pipelines, and the utilization of multimodal data sources to improve generalizability and clinical utility.

REFERENCES

- [1] Ch. Vasavi and G. Sujatha, "Data-driven heart disease prediction: A comparative study of ML models," in *Proceedings of the 4th International Conference on Innovative Mechanisms for Industry Applications (ICIMIA)*, IEEE, 2025, pp. 1711–1716, doi: 10.1109/ICIMIA67127.2025.11200983.
- [2] M. Ahmed, R. Hassan, S. Datto, S. Saleh, S. Islam, K. Redwan, and S. Mahmood, "A comparative study of machine learning models for cardiovascular risk prediction using PCA-transformed Framingham data," in *Proceedings of the International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)*, IEEE, 2025, doi: 10.1109/QPAIN66474.2025.11171785.
- [3] R. J. Roopini, D. B. G. Deepa, and P. N. G. Pooja, "Assessing and exploring machine learning techniques for cardiovascular disease prediction using Cleveland and Framingham datasets," in *Proceedings of the 2nd International Conference on Network, Multimedia and Information Technology (NMITCON)*, IEEE, 2024, doi: 10.1109/NMITCON62075.2024.10698965.
- [4] K. Kayalvizhi, S. Kanchana, S. P. Silvia, C. Padmavathy, B. S. N. V. R. Murthy, and V. Thangavel, "Optimized CNN-BiLSTM with attention: A high performance model for predicting heart disease using Cleveland and Framingham datasets," *Journal of Machine and Computing*, vol. 4, no. 4, pp. 1195–1208, 2024, doi: 10.53759/7669/jmc202404110.
- [5] E. Yuda, I. Kaneko, and D. Hirahara, "Machine-learning insights from the Framingham Heart Study: Enhancing cardiovascular risk prediction and monitoring," *Applied Sciences*, vol. 15, no. 15, p. 8671, 2025, doi: 10.3390/app15158671.
- [6] T. O. Omotehinwa, D. O. Oyewola, and E. G. Mounq, "Optimizing the light gradient-boosting machine algorithm for an efficient early detection of coronary heart disease," *Informatics and Health*, vol. 1, pp. 70–81, 2024, doi: 10.1016/j.infoh.2024.06.001.
- [7] E. Yuda, I. Kaneko, and D. Hirahara, "Designing algorithms for wearable sensors: Insights from the Framingham Heart Study dataset for enhanced cardiovascular health monitoring," *Preprints*, 2025, doi: 10.20944/preprints202502.2135.v1.
- [8] A. Bhardwaj, "Framingham Heart Study Dataset," Kaggle, 2022. [Online]. Available: <https://www.kaggle.com/datasets/aasheesh200/framingham-heart-study-dataset>
- [9] D. Dua and C. Graff, "UCI Machine Learning Repository: Heart Disease Data Set (Cleveland)," University of California, Irvine, 2019. [Online]. Available: <https://archive.ics.uci.edu/ml/datasets/Heart+Disease>
- [10] World Health Organization, "Cardiovascular diseases (CVDs)," World Health Organization, Geneva, Switzerland, Jul. 31, 2025. [Online]. Available: [https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-\(cvds\)](https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds))